

Aspects of Robust Mean-Variance Hedging

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1 Introduction

Suppose we are in a market, where we can trade financial securities or derivatives freely without constraints. Prices of objects in the market are not constant, that is to say they vary with the time. We call a person an investor, if he is active in trading with the expectation of receiving financial returns. When a derivative is traded, the risk-averse nature of an investor often requires some hedging strategy, which is an instrument to reduce the losses while adverse price movements happen. This can be usually done by constructing a portfolio, which, in short, is a combination of financial assets. We may also change share of the assets in portfolio according to the market situation. If we know what the true model of the market is, we can often find a good strategy to minimize the risk by a certain given initial cost. In the reality however, we usually do not know the true model. Instead, we have a few candidates for the true model. Finding a good hedging strategy in this case is more difficult. A good strategy in one model may act very badly in another one. Instead of observing the models separately, we are interested in a strategy, which minimizes the largest possible loss.

We study the mean-variance hedging problem with the model risk in a discrete time model. Time is discrete means that we can only observe prices of the assets at some certain given dates and we can only trade them at those dates. Matsumoto [1] has already studied the case of a one-period model with one risky asset. He observes a certain candidate set of market models, defines the minimum hedging error and proves the existence of an optimal strategy in this case. After that, he presents a way to calculate this strategy and gives some numerical examples. In this thesis, we will focus on the existence part of the problem but in some more difficult settings.

An overview of the thesis is as follows. In Section 2 we introduce the problem we study, which is a generalisation of the problem introduced by Matsumoto [1]. His approach to prove the existence of an optimal strategy is also reviewed here. In Section 3, Sion's Minimax Theorem is introduced, which is the main instrument we use to get the desired result. In Section 4 to 7 we show applications of Sion's Minimax Theorem under different model assumptions to prove the existence result. Finally, in Section 8 we generalize the candidate set and try receiving a similar result in that case.

2 Overview of the problem and approach of Matsumoto

2.1 General setup

In this section, we will introduce the general problem of research and aim of this thesis.

Let us consider a complete probability space $(\Omega, \mathcal{F}, P)$. Here P is a probability measure, which we call the true model. Let $\mathcal{Q}$ be a family of probability measures on $(\Omega, \mathcal{F})$, which will be specified later in some concrete cases. We denote the expectation under $Q \in \mathcal{Q}$ by $E^Q[\cdot]$. Let us consider the situation, that our investor does not know the true model P , but he knows that P belongs to the family $\mathcal{Q}$. This means that

$$P \in \mathcal{Q}.$$

In other words, $\mathcal{Q}$ is the set of all candidates for the true model.

We consider a T -period case between time 0 and T . This means, we can observe prices of assets and make trading decisions from time 0 to $T - 1$. Suppose there is one non-risky asset and n risky assets in the market. An asset is called non-risky, if its price at any time t is already known at time $t - 1$, while we can only know the time- t -price of a risky asset at time t . The value of the risky assets at time t is denoted by S_t . Note that

$$S_t = \begin{pmatrix} S_t^1 \\ S_t^2 \\ \vdots \\ S_t^n \end{pmatrix}$$

is a multivariate random variable taking values in $\mathbb{R}^n$. Here S_t^i denotes the price of the i -th risky asset at time t . We also set price of the non-risky asset to be S_t^0 and use the convention $S_0^0 = 1$. Let us consider a filtration $\mathbb{F} := (\mathcal{F}_t)_{t=0,1,\dots,T}$. This filtration represents the information structure. We also assume that $\mathcal{F}_0$ is P -trivial and $\mathcal{F}_T = \mathcal{F}$, which means there is no information available at the beginning and everything is known at the maturity T . The price process $(S_t)_{t=0,1,\dots,T}$ is then assumed to be $\mathbb{F}$ -adapted and $(S_t^0)_{t=1,2,\dots,T}$ to be $\mathbb{F}$ -predictable. In particular, this means that S_0 is a P -almost surely constant vector.

Let us define $\mathcal{A}(\mathbb{R}^n)$ to be the set of all $\mathbb{F}$ -predictable and $\mathbb{R}^n$ -valued random processes. Let $\Pi \subseteq \mathcal{A}(\mathbb{R}^n)$ be the set of all admissible strategies, which will be stip-

ulated later in some concrete cases. We recall the definition of *portfolio* introduced by Possamaï [2, Section 2]. Some adjustments of the definition are made to fit our setup.

Definition 2.1. A *portfolio* is an $\mathbb{F}$ -predictable process $(\eta, \pi) := \{(\eta_t, \pi_t) : t \in \{1, \dots, T\}\}$ with values in $\mathbb{R} \times \mathbb{R}^n$, where n is the number of risky assets. More precisely, for any $t \in \{1, \dots, T\}$, η_t represents the number of shares of the non-risky asset contained in the portfolio at time $t - 1$, and for any $i \in \{1, \dots, n\}$, π_t^i represents the number of shares of asset i contained in the portfolio at time $t - 1$. The wealth associated to such a portfolio at any time t is thus

$$\eta_{t+1}S_t^0 + \pi_{t+1} \cdot S_t, \quad t \in \{0, \dots, T - 1\},$$

where $\eta_{t+1}S_t^0$ is wealth in the non-risky asset and $\pi_{t+1} \cdot S_t$ is wealth in the risky assets. For any portfolio (η, π) , we now define the associated *consumption process* $c = \{c_t : t \in \{1, \dots, T - 1\}\}$ by

$$c_t := -\eta_{t+1}S_t^0 - \pi_{t+1} \cdot S_t + (\eta_t S_t^0 + \pi_t \cdot S_t).$$

c_t is therefore the difference between the value of the portfolio at time t before the investor rebalances his positions from (η_t, π_t) to (η_{t+1}, π_{t+1}) . If $c_t \geq 0$, then the investor has consumed some of his wealth at time t , and if $c_t \leq 0$, then the investor has received an income from a source outside of the market. A portfolio is said to be *self-financing* if its consumption process c is always equal to 0.

We also recall the notion of *discounted value* from Possamaï [2, Section 2].

Definition 2.2. For an arbitrary process $V := \{V_t : t \in \{0, \dots, T\}\}$ we define its *discounted value* $\tilde{V} := \{\tilde{V}_t : t \in \{0, \dots, T\}\}$ by

$$\tilde{V}_t := \frac{V_t}{S_t^0}, \quad t \in \{0, \dots, T\}.$$

In the rest of paper, if nothing else is stated, all prices are discounted by the non-risky asset.

Remark 2.3. We fix a starting capital c . One can show that for any investing strategy $\pi \in \Pi$, the corresponding wealth of a self-financing portfolio at maturity T is given by

$$X_T^{c, \pi} = c + \sum_{t=1}^T \pi_t \cdot (S_t - S_{t-1}). \quad (2.1)$$

We want to use such a portfolio to hedge a derivative whose payoff is H . In fact H is a random variable on $(\Omega, \mathcal{F})$, whose output is the price of the derivative at maturity T . Now we define the hedging error under a strategy $\pi \in \Pi$ and a model $Q \in \mathcal{Q}$ to be

$$L(\pi, Q) = E^Q[(H - X_T^{c, \pi})^2], \quad \pi \in \Pi, \quad Q \in \mathcal{Q}. \quad (2.2)$$

Remember the initial capital c is fixed in our case. We aim to find an optimal strategy π^* which minimizes the hedging error in the worst possible situation, if such a strategy exists. In other words, we want to minimize the function

$$\pi \mapsto \sup_{Q \in \mathcal{Q}} L(\pi, Q)$$

over all $\pi \in \Pi$. We also define the minimum hedging error L^* to be

$$L^* = \inf_{\pi \in \Pi} \sup_{Q \in \mathcal{Q}} L(\pi, Q).$$

In particular, if the infimum is attained at some optimal strategy π^* , we also have

$$L^* = \sup_{Q \in \mathcal{Q}} L(\pi^*, Q).$$

In general, it is uncertain if the infimum and the supremum above are attained. The main topic in this paper is to discuss the existence of such an optimal strategy π^* in different situations and under various assumptions.

2.2 Matsumoto's approach

This subsection is generally based on Matsumoto [1]. Here a repetition of his approach is given.

We consider a one-period model between time 0 and 1. Suppose there is only one risky asset in our market. This means that $T = 1$ and $n = 1$. For $M \in \mathbb{N}$, let $Q_1, Q_2, \dots, Q_M$ be probability measures. For $1 \leq m \leq M$, we call Q_m a base model. The candidate set $\mathcal{Q}$ of probability measures is chosen to be

$$\mathcal{Q} = \left\{ Q = \sum_{m=1}^M w_m Q_m : (w_1, w_2, \dots, w_M) \in \mathcal{W} \right\}, \quad (2.3)$$

where

$$\mathcal{W} = \left\{ w = (w_1, w_2, \dots, w_M) \in \mathbb{R}^M : 0 \leq w_m \leq 1, \sum_{m=1}^M w_m = 1 \right\}. \quad (2.4)$$

In this setting, we choose the set of all admissible strategies as $\Pi = \mathbb{R}$. Let us denote $X = S_1 - S_0$. We assume that the random variable X satisfies

$$0 < E^{Q_m}[X^2] < \infty, \quad \forall 1 \leq m \leq M.$$

This also implies that for all $Q = \sum_{m=1}^M w_m Q_m \in \mathcal{Q}$, we have

$$0 < E^Q[X^2] < \infty \tag{2.5}$$

since

$$E^Q[X^2] = \sum_{m=1}^M w_m E^{Q_m}[X^2].$$

We also assume that the payoff H of the derivative is square-integrable for $\mathcal{Q}$, in the sense that

$$E^{Q_m}[H^2] < \infty, \quad \forall 1 \leq m \leq M$$

or equivalently

$$E^Q[H^2] < \infty, \quad \forall Q \in \mathcal{Q}.$$

For a fixed initial capital c and any $\pi \in \mathbb{R}$, the value of the hedging portfolio is then

$$c + \pi X.$$

The hedging error or loss function in this context is then

$$L(\pi, Q) = E^Q[(H - c - \pi X)^2] \tag{2.6}$$

$$= E^Q[X^2]\pi^2 - 2E^Q[\bar{H}X]\pi + E^Q[\bar{H}^2], \tag{2.7}$$

where

$$\bar{H} = H - c.$$

For simplicity, we set for $1 \leq m \leq M$

$$L_m(\pi) = L(\pi, Q_m).$$

In this simple setting, Matsumoto [1, Theorem 3.1] proves the existence of an optimal strategy by a direct calculation.

Theorem 2.4. *In the above setting, there exists a unique optimal strategy, and the minimum hedging error is finite.*

Proof. We have

$$\begin{aligned}
\sup_{Q \in \mathcal{Q}} L(\pi, Q) &= \sup_{Q \in \mathcal{Q}} E^Q[(\bar{H} - \pi X)^2] \\
&= \sup_{(w_1, \dots, w_M) \in \mathcal{W}} \sum_{m=1}^M w_m E^{Q_m}[(\bar{H} - \pi X)^2] \\
&= \max_{1 \leq m \leq M} E^{Q_m}[(\bar{H} - \pi X)^2] \\
&= \max_{1 \leq m \leq M} L_m(\pi).
\end{aligned}$$

Here, the third equality is true because the supremum over a convex set in $\mathbb{R}$ generated by a finite number of points is always attained as a maximum over the extremal points.

Define

$$\bar{L}(\pi) = \max_{1 \leq m \leq M} L_m(\pi).$$

Since we know that

$$L_m(\pi) = E^{Q_m}[X^2]\pi^2 - 2E^{Q_m}[\bar{H}X]\pi + E^{Q_m}[\bar{H}^2]$$

and the coefficient of the squared term π^2 is positive by assumption (2.5), the function $\pi \mapsto L_m(\pi)$ is strictly convex and continuous for all $1 \leq m \leq M$. Every L_m also has the property

$$\lim_{|\pi| \rightarrow \infty} L_m(\pi) = +\infty.$$

As a maximum of finitely many such functions, $\bar{L}$ is hence also strictly convex, continuous and has the property

$$\lim_{|\pi| \rightarrow \infty} \bar{L}(\pi) = +\infty.$$

We choose a sufficiently large real number N with $N > \bar{L}(0)$. There exists $\delta > 0$ such that if π satisfies $|\pi| > \delta$, then

$$\bar{L}(\pi) > N.$$

Since $\bar{L}$ is strictly convex and continuous, there exists a unique $\pi_0 \in [-\delta, \delta]$ satisfying

$$\bar{L}(\pi_0) = \min_{|\pi| \leq \delta} \bar{L}(\pi).$$

For any $|\pi| > \delta$, we have

$$\bar{L}(\pi_0) \leq \bar{L}(0) < N < \bar{L}(\pi). \quad (2.8)$$

Therefore, π_0 is the minimum of $\bar{L}$ over all $\pi \in \mathbb{R}$ and we get

$$L^* = \inf_{\pi \in \mathbb{R}} \sup_{Q \in \mathcal{Q}} L(\pi, Q) = \inf_{\pi \in \mathbb{R}} \bar{L}(\pi) = \bar{L}(\pi_0) = \sup_{Q \in \mathcal{Q}} L(\pi_0, Q) < N. \quad (2.9)$$

We have thus proved that π_0 is a unique optimal strategy and that the minimum hedging error L^* is finite. $\square$

Matsumoto [1, Section 3] also proves that $L^* = \max_{Q \in \mathcal{Q}} \min_{\pi \in \mathbb{R}} L(\pi, Q)$ in this special case, that is, we can change the order of min and max in this situation. He also gives some numerical examples showing how to find the optimal strategy. Since we are only interested in the existence of such a strategy here, we omit the proofs and refer to his paper.

3 The minimax theorem of Sion

In the previous chapter, we have seen the existence of an optimal strategy in a one-period model with one risky asset and a particular set $\mathcal{Q}$. In this chapter, we introduce Sion's minimax theorem, which allows us to extend the model such that it contains more periods or more risky assets. It also allows us to consider more general candidate set $\mathcal{Q}$.

Before stating the minimax theorem, let us recall some definitions we use in this context.

Definition 3.1. Let S be a convex subset of a real vector space and $f : S \rightarrow \mathbb{R}$ a function. Then f is called *quasi-convex* if

$$f(\lambda x + (1 - \lambda)y) \leq \max\{f(x), f(y)\} \quad (3.1)$$

for all $x, y \in S$ and $\lambda \in [0, 1]$. If " $\leq$ " is replaced by " $\geq$ " and " $\max$ " by " $\min$ " in (3.1), f is called *quasi-concave*.

Remark 3.2. Note that if $f : S \rightarrow \mathbb{R}$ is convex, then it is also quasi-convex since for all $x, y \in S$ and $\lambda \in [0, 1]$, we have

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \leq \max\{f(x), f(y)\}.$$

Similarly, if $f : S \rightarrow \mathbb{R}$ is concave, then it is also quasi-concave.

Definition 3.3. Let S be a topological space and $f : S \rightarrow \mathbb{R}$ a real valued function. Let x_0 be a point in S . The function f is said to be *upper semi-continuous at point* x_0 if for every $y \in \mathbb{R}$ with $y > f(x_0)$ there exists a neighborhood U of x_0 such that $y > f(u)$ for all $u \in U$. If " $>$ " is replaced by " $<$ " above, f is said to be *lower semi-continuous at point* x_0 .

We say f is lower/upper semi-continuous on S if it is lower/upper semi-continuous at every point of S .

Remark 3.4. A function $f : S \rightarrow \mathbb{R}$ is continuous on S if and only if it is lower semi-continuous and upper semi-continuous on S .

We are now at the step to introduce the minimax theorem of Sion as referenced by Komiya [3].

Theorem 3.5 (Sion's minimax theorem). *Let X be a compact convex subset of a linear topological space and Y a convex subset of a linear topological space. If f is a real-valued function on $X \times Y$ such that*

$f(x, \cdot)$ is quasi-convex and lower semi-continuous on Y , $\forall x \in X$, and
 $f(\cdot, y)$ is quasi-concave and upper semi-continuous on X , $\forall y \in Y$,
then we have

$$\max_{x \in X} \inf_{y \in Y} f(x, y) = \inf_{y \in Y} \max_{x \in X} f(x, y).$$

Remark 3.6. If one manages to show that for any fixed $x \in X$, $\inf_{y \in Y} f(x, y)$ is attained, then the upper semi-continuity of the function $f(\cdot, y)$ also ensures the existence of $\max_{x \in X} \min_{y \in Y} f(x, y)$. In fact, since $f(\cdot, y)$ is upper semi-continuous for all $y \in Y$, we have that $\inf_{y \in Y} f(\cdot, y) = \min_{y \in Y} f(\cdot, y)$ is upper semi-continuous as well. Hence, $\sup_{x \in X} \min_{y \in Y} f(x, y) = \max_{x \in X} \min_{y \in Y} f(x, y)$. Note that X is compact by assumption. By applying Sion's minimax theorem we get then

$$\max_{x \in X} \min_{y \in Y} f(x, y) = \min_{y \in Y} \max_{x \in X} f(x, y)$$

In particular, there exists an optimal $y^* \in Y$, such that

$$\max_{x \in X} f(x, y^*) = \min_{y \in Y} \max_{x \in X} f(x, y)$$

Sion's minimax theorem suggests us to choose the space of strategies Π as Y and the family of probability measures $\mathcal{Q}$ as X , since we want to minimize the loss function L in the argument π and maximize it in the argument Q .

In the following chapters, detailed applications of the minimax theorem in different settings and situations will be shown.

4 Application of Sion's minimax Theorem in the case of Matsumoto

In this section, we give an alternative proof of the existence part of Theorem 2.4 by using Sion's Minimax theorem (Theorem 3.5).

Suppose we are in the same setting as section 2.2. The idea is that we want to apply Theorem 3.5 to the loss function L . As we have already mentioned in the previous chapter, we want to choose the space of strategies Π as Y and the family of probability measures $\mathcal{Q}$ as X . Since we choose $\Pi = \mathbb{R}$ in this setting, it is natural to choose the standard euclidean topology on $\mathbb{R}$. The space Π satisfies therefore the conditions in Theorem 3.5, namely Π is a convex subset of a linear topological space.

In order to analyse the set $\mathcal{Q}$, we first recall the notion of signed measures referenced from Bell [4, Section 2].

Definition 4.1. Let $\mathcal{F}$ be a σ -algebra on a set X and let $E \in \mathcal{F}$. A countable collection $\{E_i\} \subseteq \mathcal{F}$ whose members are pairwise disjoint and that satisfies the property $E = \bigcup_{i=1}^{\infty} E_i$ is called a *partition* of E . A *signed measure* is a function $\mu : \mathcal{F} \rightarrow \mathbb{R}$ such that if $E \in \mathcal{F}$, then

$$\mu(E) = \sum_{i=1}^{\infty} \mu(E_i)$$

for every partition $\{E_i\}$ of E . We denote by $ca(\mathcal{F})$ the set of signed measures on $\mathcal{F}$.

Remark 4.2. Jordan decomposition Theorem tells us that every signed measure μ defined on $\mathcal{F}$ has a unique decomposition into a difference $\mu = \mu^+ - \mu^-$, where μ^+ and μ^- are positive measures. For any $\mathcal{F}$ -measurable function f , one can thus define the integration of f respect to the signed measure μ to be

$$\int f d\mu = \int f d\mu^+ - \int f d\mu^-.$$

One result we want to use later is that for any $\mathcal{F}$ -measurable function f , $\alpha, \beta \in \mathbb{R}$ and $\mu_1, \mu_2 \in ca(\mathcal{F})$ we have

$$\int f d(\alpha\mu_1 + \beta\mu_2) = \alpha \int f d\mu_1 + \beta \int f d\mu_2.$$

The notion of signed measure is a generalisation of the term measure since a signed measure takes values in $\mathbb{R}$ whereas a measure only takes non-negative values.

It is important here to consider the space of signed measures because we want a linear topological space in the background. One can show that the space $ca(\mathcal{F})$ is actually a vector space.

It is clear by (2.3) and (2.4) that the set $\mathcal{Q}$ we observe here is convex. In fact, it is just the convex hull of the basic models $Q_1, Q_2, \dots, Q_M$. Though the space $ca(\mathcal{F})$ is infinite dimensional, we can reduce the problem to a finite dimensional one by restricting to the space V spanned by the measures $Q_1, Q_2, \dots, Q_M$. This means

$$V = \left\{ \sum_{m=1}^M \alpha_m Q_m : \alpha_m \in \mathbb{R} \right\}. \quad (4.1)$$

We can then identify every $Q \in V$ with a unique M -dimensional vector in $\mathbb{R}^M$. Thus, let us define the distance between any two signed measures Q and Q' to be the euclidean distance between the appropriate vectors. With this consideration, we identify the set $\mathcal{Q}$ with a simplex in $\mathbb{R}^M$, which is clearly compact. Hence, we have that $\mathcal{Q}$ is a compact convex subset of the linear topological space V .

We fix a measure $Q \in \mathcal{Q}$ and observe the function

$$\begin{aligned} L_Q : \mathbb{R} &\rightarrow \mathbb{R} \\ \pi &\mapsto L(\pi, Q). \end{aligned}$$

Proposition 4.3. *For any $Q \in \mathcal{Q}$, the function L_Q is lower semi-continuous and quasi-convex. Furthermore, $\inf_{\pi \in \mathbb{R}} L_Q(\pi)$ is attained.*

Proof. let us fix an arbitrary $Q \in \mathcal{Q}$. We know from equation (2.7) and assumption (2.5) that

$$L_Q(\pi) = E^Q[X^2]\pi^2 - 2E^Q[\bar{H}X]\pi + E^Q[\bar{H}^2].$$

This is a quadratic function in π , and the coefficient before π^2 is strictly positive. Thus the function $L_Q(\cdot) = L(\cdot, Q)$ is continuous and strictly convex. By Remark 3.4 and Remark 3.2, $L(\cdot, Q)$ is also lower semi-continuous and quasi-convex. From this expression it is also clear to see that $\inf_{\pi \in \mathbb{R}} L_Q(\pi)$ is attained. $\square$

For any fixed $\pi \in \mathbb{R}$, we observe the function

$$\begin{aligned} \tilde{L}_\pi : V &\rightarrow \mathbb{R} \\ Q &\mapsto \int_{\Omega} (\bar{H} - \pi X)^2 dQ. \end{aligned}$$

We can show that the function $\tilde{L}_\pi$ is linear in Q . In fact, for all $Q_1, Q_2 \in V$ and

$\alpha, \beta \in \mathbb{R}$ we have

$$\begin{aligned}
\tilde{L}_\pi(\alpha Q_1 + \beta Q_2) &= \int_{\Omega} (\bar{H} - \pi X)^2 d(\alpha Q_1 + \beta Q_2) \\
&= \alpha \int_{\Omega} (\bar{H} - \pi X)^2 dQ_1 + \beta \int_{\Omega} (\bar{H} - \pi X)^2 dQ_2 \\
&= \alpha \tilde{L}_\pi(Q_1) + \beta \tilde{L}_\pi(Q_2).
\end{aligned} \tag{4.2}$$

As a linear function over a finite dimensional vector space, we have thus $\tilde{L}_\pi$ is continuous. We now observe the restriction of $\tilde{L}_\pi$ on the subset $\mathcal{Q}$, which is denoted by L_π . Namely, we have

$$\begin{aligned}
L_\pi : \mathcal{Q} &\rightarrow \mathbb{R} \\
Q &\mapsto \int_{\Omega} (\bar{H} - \pi X)^2 dQ = E^Q[(\bar{H} - \pi X)^2] = L(\pi, Q).
\end{aligned}$$

Proposition 4.4. *For any $\pi \in \mathbb{R}$, the function L_π is upper semi-continuous and quasi-concave.*

Proof. Let us fix an arbitrary $\pi \in \mathbb{R}$. As a restriction of a continuous function, we have that $L_\pi(\cdot) = L(\pi, \cdot)$ is also continuous, hence also upper semi-continuous by Remark 3.4. Furthermore, for any $\lambda \in [0, 1]$ and any $Q_1, Q_2 \in \mathcal{Q}$ we have

$$\begin{aligned}
L_\pi(\lambda Q_1 + (1 - \lambda)Q_2) &= E^{\lambda Q_1 + (1 - \lambda)Q_2}[(\bar{H} - \pi X)^2] \\
&= \lambda E^{Q_1}[(\bar{H} - \pi X)^2] + (1 - \lambda) E^{Q_2}[(\bar{H} - \pi X)^2] \\
&= \lambda L_\pi(Q_1) + (1 - \lambda) L_\pi(Q_2) \\
&\geq \min\{L_\pi(Q_1), L_\pi(Q_2)\}.
\end{aligned} \tag{4.3}$$

Thus we have shown that the function L_π is quasi-concave. $\square$

We now provide an alternative proof of the existence of an optimal strategy in Theorem 2.4.

Proof. We know that $\Pi = \mathbb{R}$ is a convex subset of a linear topological space and $\mathcal{Q}$ is a compact and convex subset of the linear topological space V . The function

$$\begin{aligned}
L : \Pi \times \mathcal{Q} &\rightarrow \mathbb{R} \\
(\pi, Q) &\mapsto E^Q[(\bar{H} - \pi X)^2]
\end{aligned}$$

is a real-valued function on $\Pi \times \mathcal{Q}$, which satisfies

$L(\cdot, Q)$ is quasi-convex and lower semi-continuous on Π , $\forall Q \in \mathcal{Q}$, and

$L(\pi, \cdot)$ is quasi-concave and upper semi-continuous on $\mathcal{Q}$, $\forall \pi \in \Pi$.
We can then apply Sion's minimax theorem (Theorem 3.5) to show

$$\max_{Q \in \mathcal{Q}} \inf_{\pi \in \Pi} L(\pi, Q) = \inf_{\pi \in \Pi} \max_{Q \in \mathcal{Q}} L(\pi, Q).$$

Furthermore, we have also shown that for any fixed $Q \in \mathcal{Q}$, $\inf_{\pi \in \Pi} L(\pi, Q)$ is attained. We can then apply Remark 3.6 to show the existence of an optimal strategy $\pi^* \in \Pi$, such that

$$\max_{Q \in \mathcal{Q}} L(\pi^*, Q) = \min_{\pi \in \Pi} \max_{Q \in \mathcal{Q}} L(\pi, Q).$$

This is just the desired result. □

5 The one-period, n-risky-asset case

In the following chapters, we would like to extend the model and discuss the existence of optimal strategy in more general situations.

In this chapter we will focus on the one-period, n-risky-asset case, that is to say we have more risky assets or stocks to trade. As in the previous sections, we still observe the candidate set $\mathcal{Q}$ introduced by Matsumoto in (2.3). We also assume that all strategies are admissible in our case, that is to say

$$\Pi = \mathbb{R}^n = \mathcal{A}(\mathbb{R}^n).$$

The second equality is true since we only observe a one-period model in this case. By equation (2.1), value of the hedging portfolio at time 1 is

$$X^{c,\pi} = c + \pi \cdot X,$$

where $X = S_1 - S_0$ is a random vector taking values in $\mathbb{R}^n$ in our case. Denote

$$X = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix}.$$

We make some similar assumptions to the ones in Section 2.2. Let us assume

$$0 < E^Q[X_i^2] < \infty, \quad \forall i \in \{1, \dots, n\}, \quad \forall Q \in \mathcal{Q} \quad (5.1)$$

and

$$E^Q[H^2] < \infty, \quad \forall Q \in \mathcal{Q}. \quad (5.2)$$

5.1 Proof of the existence under closedness assumption

Let us define the set $\mathcal{G}$ of end values as follows

$$\mathcal{G} = \{\pi \cdot X : \pi \in \mathbb{R}^n\}.$$

Recall that $L^2(Q)$ denotes the space of all square-integrable functions defined on the set Ω respect to the measure Q . Equation (5.1) and (5.2) just tell us that

$$X_i \in L^2(Q), \quad \forall i \in \{1, \dots, n\}, \quad \forall Q \in \mathcal{Q}$$

and

$$H \in L^2(Q), \quad \forall Q \in \mathcal{Q}.$$

Since $L^2(Q)$ is a vector space, we also get that

$$\mathcal{G} = \{\pi \cdot X : \pi \in \mathbb{R}^n\} \subseteq L^2(Q), \quad \forall Q \in \mathcal{Q}.$$

Proposition 5.1. *For all $Q \in \mathcal{Q}$, the set $\mathcal{G}$ is actually a subspace of $L^2(Q)$.*

Proof. For any $G_1, G_2 \in \mathcal{G}$ we have

$$G_1 = \pi_1 \cdot X$$

and

$$G_2 = \pi_2 \cdot X$$

for some $\pi_1, \pi_2 \in \mathbb{R}^n$. Thus

$$G_1 + G_2 = (\pi_1 + \pi_2) \cdot X$$

is also an element of $\mathcal{G}$. Similarly, for any $\alpha \in \mathbb{R}$ we have

$$\alpha G_1 = \alpha \pi_1 \cdot X$$

is an element of $\mathcal{G}$. Hence, $\mathcal{G}$ is itself a vector space. □

Let us assume in addition that $\mathcal{G}$ is a closed subspace of $L^2(Q)$ for all $Q \in \mathcal{Q}$.

Again, we want to verify that all conditions of the Sion's Minimax Theorem are satisfied and we can apply it in this situation to prove the existence of an optimal strategy.

We choose the euclidean topology on $\Pi = \mathbb{R}^n$ and the same topology on $\mathcal{Q}$ as in section 4. For any fixed measure $Q \in \mathcal{Q}$ we observe the function

$$\begin{aligned} L_Q : \mathbb{R}^n &\rightarrow \mathbb{R} \\ \pi &\mapsto L(\pi, Q). \end{aligned}$$

Proposition 5.2. *For any $Q \in \mathcal{Q}$, the function L_Q is lower semi-continuous and quasi-convex.*

Proof. We fix an arbitrary $Q \in \mathcal{Q}$. Recall that by equation (2.2) we have in this case

$$L(\pi, Q) = E^Q[(H - c - \pi \cdot X)^2] = E^Q[(\bar{H} - \pi \cdot X)^2], \quad (5.3)$$

where $\bar{H} = H - c$. Note that for all $Q \in \mathcal{Q}$, we also have

$$\bar{H} \in L^2(Q) \quad (5.4)$$

again since $L^2(Q)$ is a vector space. We also expand formula (5.3) above

$$\begin{aligned} L(\pi, Q) &= E^Q[(\bar{H} - \pi \cdot X)^2] \\ &= E^Q[(\bar{H} - \sum_{i=1}^n \pi_i X_i)^2] \\ &= E^Q[\bar{H}^2] - 2 \sum_{i=1}^n E^Q[\pi_i X_i \bar{H}] + 2 \sum_{i \neq j} E^Q[\pi_i \pi_j X_i X_j] + \sum_{i=1}^n E^Q[(\pi_i X_i)^2] \\ &= E^Q[\bar{H}^2] - 2 \sum_{i=1}^n E^Q[X_i \bar{H}] \pi_i + 2 \sum_{i \neq j} E^Q[X_i X_j] \pi_i \pi_j + \sum_{i=1}^n E^Q[X_i^2] \pi_i^2. \end{aligned} \quad (5.5)$$

By (5.1), (5.4) and application of Hölder Inequality, all expectations above are finite. From this expansion it is clear to see that the function L_Q is a multidimensional polynomial. One direct result is that L_Q is continuous, hence also lower semi-continuous by Remark 3.4. In addition, for all $\lambda \in [0, 1]$ and $\pi, \pi' \in \mathbb{R}^n$ we have

$$\begin{aligned} L_Q(\lambda\pi + (1-\lambda)\pi') &= E^Q\left[(\bar{H} - (\lambda\pi + (1-\lambda)\pi') \cdot X)^2\right] \\ &= E^Q[(\lambda(\bar{H} - \pi \cdot X) + (1-\lambda)(\bar{H} - \pi' \cdot X))^2] \\ &\leq E^Q[\lambda(\bar{H} - \pi \cdot X)^2 + (1-\lambda)(\bar{H} - \pi' \cdot X)^2] \\ &= \lambda E^Q[(\bar{H} - \pi \cdot X)^2] + (1-\lambda) E^Q[(\bar{H} - \pi' \cdot X)^2] \\ &= \lambda L_Q(\pi) + (1-\lambda) L_Q(\pi'), \end{aligned}$$

where we use the fact that the function $f(x) = x^2$ is convex at the inequality above. From this calculation we see that the function L_Q is convex. By Remark 3.2, it is also quasi-convex. □

Proposition 5.3. *For any candidate measure $Q \in \mathcal{Q}$, the function L_Q attains its infimum.*

Proof. Let Q be an arbitrary candidate measure in $\mathcal{Q}$. Since by our assumption the subspace $\mathcal{G}$ is closed in $L^2(Q)$, we can observe the projection of $\bar{H}$ on the subspace $\mathcal{G}$. Let us denote this projection by $P(\bar{H})$. We can then use the property of linear

projection to get

$$E^Q[(\bar{H} - P(\bar{H}))^2] \leq E^Q[(\bar{H} - G)^2], \quad \forall G \in \mathcal{G}. \quad (5.6)$$

Since $P(\bar{H}) \in \mathcal{G}$, there exists a $\bar{\pi} \in \mathbb{R}^n$ such that

$$P(\bar{H}) = \bar{\pi} \cdot X.$$

Thus we get

$$L_Q(\bar{\pi}) = E^Q[(\bar{H} - \bar{\pi} \cdot X)^2] = \min_{\pi \in \mathbb{R}^n} E^Q[(\bar{H} - \pi \cdot X)^2] = \min_{\pi \in \mathbb{R}^n} L_Q(\pi). \quad (5.7)$$

In fact, we can show equation (5.7) by contradiction. Assume by contradiction there is some $\tilde{\pi} \in \mathbb{R}^n$ such that $L_Q(\tilde{\pi}) < L_Q(\bar{\pi})$. In other words, we have

$$E^Q[(\bar{H} - \tilde{\pi} \cdot X)^2] < E^Q[(\bar{H} - \bar{\pi} \cdot X)^2].$$

If we define $\tilde{G} := \tilde{\pi} \cdot X$, the inequality above becomes to

$$E^Q[(\bar{H} - \tilde{G})^2] < E^Q[(\bar{H} - P(\bar{H}))^2].$$

Since $\tilde{G} \in \mathcal{G}$ by definition, this contradicts (5.6). Hence we conclude that equation (5.7) is true. In other words, the function L_Q attains its infimum at some $\bar{\pi} \in \mathbb{R}^n$. $\square$

Let V be defined as in (4.1). For any fixed $\pi \in \mathbb{R}^n$, we observe the function

$$\begin{aligned} \tilde{L}_\pi : V &\rightarrow \mathbb{R} \\ Q &\mapsto \int_{\Omega} (\bar{H} - \pi \cdot X)^2 dQ. \end{aligned}$$

We can do almost the same calculation as in (4.2) to show that the function $\tilde{L}_\pi$ is linear, thus also continuous as a linear function over a finite dimensional vector space. We also observe the restriction of $\tilde{L}_\pi$ on the subset $\mathcal{Q}$

$$\begin{aligned} L_\pi : \mathcal{Q} &\rightarrow \mathbb{R} \\ Q &\mapsto \int_{\Omega} (\bar{H} - \pi \cdot X)^2 dQ = E^Q[(\bar{H} - \pi \cdot X)^2] = L(\pi, Q). \end{aligned}$$

Proposition 5.4. *For any $\pi \in \mathbb{R}^n$, the function L_π is upper semi-continuous and quasi-concave.*

Proof. We fix a vector $\pi \in \mathbb{R}^n$. As restriction of a continuous function, L_π is also

continuous and hence also upper semi-continuous by Remark 3.4. For any $\lambda \in [0, 1]$ and any $Q_1, Q_2 \in \mathcal{Q}$ we have

$$\begin{aligned}
L_\pi(\lambda Q_1 + (1 - \lambda)Q_2) &= E^{\lambda Q_1 + (1 - \lambda)Q_2}[(\bar{H} - \pi \cdot X)^2] \\
&= \lambda E^{Q_1}[(\bar{H} - \pi \cdot X)^2] + (1 - \lambda) E^{Q_2}[(\bar{H} - \pi \cdot X)^2] \\
&= \lambda L_\pi(Q_1) + (1 - \lambda) L_\pi(Q_2) \\
&\geq \min\{L_\pi(Q_1), L_\pi(Q_2)\}.
\end{aligned} \tag{5.8}$$

This provides us the quasi-concavity of L_π . $\square$

We are now in the situation to prove the main theorem of this chapter.

Theorem 5.5. *In the above setting, there exists an optimal strategy $\pi^* \in \mathbb{R}^n$ such that*

$$L^* = \min_{\pi \in \mathbb{R}^n} \max_{Q \in \mathcal{Q}} L(\pi, Q) = \max_{Q \in \mathcal{Q}} L(\pi^*, Q).$$

Proof. We observe the function

$$\begin{aligned}
L : \Pi \times \mathcal{Q} &\rightarrow \mathbb{R} \\
(\pi, Q) &\mapsto E^Q[(\bar{H} - \pi \cdot X)^2],
\end{aligned}$$

where $\Pi = \mathbb{R}^n$ is a convex subset of a linear topological space and $\mathcal{Q}$ is a compact and convex subset of the linear topological V . The function L is real-valued and satisfies

$L(\cdot, Q)$ is quasi-convex and lower semi-continuous on Π , $\forall Q \in \mathcal{Q}$, and

$L(\pi, \cdot)$ is quasi-concave and upper semi-continuous on $\mathcal{Q}$, $\forall \pi \in \Pi$.

We can then apply Sion's minimax theorem (Theorem 3.5) to show

$$\max_{Q \in \mathcal{Q}} \inf_{\pi \in \Pi} L(\pi, Q) = \inf_{\pi \in \Pi} \max_{Q \in \mathcal{Q}} L(\pi, Q).$$

Furthermore, we have also shown that for any fixed $Q \in \mathcal{Q}$, $\inf_{\pi \in \Pi} L(\pi, Q)$ is attained. We can then apply Remark 3.6 to show the existence of an optimal strategy $\pi^* \in \Pi$, such that

$$\max_{Q \in \mathcal{Q}} L(\pi^*, Q) = \min_{\pi \in \Pi} \max_{Q \in \mathcal{Q}} L(\pi, Q) = L^*.$$

This is just the desired result. $\square$

5.2 Proof of the existence under uncorrelation assumption

In this subsection, we drop the assumption that the space of end values $\mathcal{G}$ is closed in $L^2(Q)$ for all $Q \in \mathcal{Q}$. Instead, we assume that all increasement of the risky assets $X_1, X_2, \dots, X_n$ are uncorrelated. Since they are risky, we also assume that

$$\text{Var}[X_i] > 0, \quad \forall i \in \{1, \dots, n\},$$

which just tells us that the risky assets are not deterministic. For any fixed $Q \in \mathcal{Q}$, we again observe the function

$$\begin{aligned} L_Q : \mathbb{R}^n &\rightarrow \mathbb{R} \\ \pi &\mapsto L(\pi, Q). \end{aligned}$$

Continuity and convexity of the function L_Q is already shown in subsection 5.1. Now we show that $\inf_{\pi \in \mathbb{R}^n} L_Q(\pi)$ is attained in our new setting by a direct calculation. Recall by (5.5) we have

$$L_Q(\pi) = E^Q[\bar{H}^2] - 2 \sum_{i=1}^n E^Q[X_i \bar{H}] \pi_i + 2 \sum_{i \neq j} E^Q[X_i X_j] \pi_i \pi_j + \sum_{i=1}^n E^Q[X_i^2] \pi_i^2.$$

This is an n -dimensional polynomial.

For sake of simplicity, we will write $E[\cdot]$ instead of $E^Q[\cdot]$ in this subsection. In order to find a minimum, we compute the gradient of L_Q , that is

$$\nabla L_Q(\pi) = \begin{pmatrix} -2E[X_1 \bar{H}] + 2 \sum_{i=1}^n E[X_1 X_i] \pi_i \\ -2E[X_2 \bar{H}] + 2 \sum_{i=1}^n E[X_2 X_i] \pi_i \\ \vdots \\ -2E[X_n \bar{H}] + 2 \sum_{i=1}^n E[X_n X_i] \pi_i \end{pmatrix}$$

We want to solve the equation

$$\nabla L_Q(\pi) = 0,$$

which is equivalent to solve

$$A\pi = v, \tag{5.9}$$

where

$$A = \begin{pmatrix} E[X_1^2] & E[X_1X_2] & \cdots & E[X_1X_n] \\ E[X_2X_1] & E[X_2^2] & \cdots & E[X_2X_n] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_nX_1] & E[X_nX_2] & \cdots & E[X_n^2] \end{pmatrix}$$

and

$$v = \begin{pmatrix} E[X_1\bar{H}] \\ E[X_2\bar{H}] \\ \vdots \\ E[X_n\bar{H}] \end{pmatrix}.$$

Since we have assumed that all X_i 's are uncorrelated, we have

$$E[X_iX_j] = E[X_i]E[X_j], \quad \forall i, j \in \{1, \dots, n\}.$$

Hence we also have

$$A = \begin{pmatrix} E[X_1^2] & E[X_1]E[X_2] & \cdots & E[X_1]E[X_n] \\ E[X_2]E[X_1] & E[X_2^2] & \cdots & E[X_2]E[X_n] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_n]E[X_1] & E[X_n]E[X_2] & \cdots & E[X_n^2] \end{pmatrix}.$$

We know that equation (5.9) has exactly one solution if the matrix A is invertible. The following Lemma provides this result.

Lemma 5.6. *In the above setting, A is invertible.*

Proof. We first observe the matrix B defined to be

$$B = \begin{pmatrix} E[X_1]^2 & E[X_1]E[X_2] & \cdots & E[X_1]E[X_n] \\ E[X_2]E[X_1] & E[X_2]^2 & \cdots & E[X_2]E[X_n] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_n]E[X_1] & E[X_n]E[X_2] & \cdots & E[X_n]^2 \end{pmatrix}$$

We claim that B is singular for $n \geq 2$. In fact, if there is some $i \in \{1, \dots, n\}$ such that $E[X_i] = 0$, then a whole row of the matrix B will be 0. B is singular in that case. We assume then without loss of generality, that $E[X_i] \neq 0$ for all $i \in \{1, \dots, n\}$. We multiply the second row of B by $E[X_1]/E[X_2]$. The new matrix $\tilde{B}$ we get is

then

$$\tilde{B} = \begin{pmatrix} E[X_1]^2 & E[X_1]E[X_2] & \cdots & E[X_1]E[X_n] \\ E[X_1]^2 & E[X_1]E[X_2] & \cdots & E[X_1]E[X_n] \\ E[X_3]E[X_1] & E[X_3]E[X_2] & \cdots & E[X_3]E[X_n] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_n]E[X_1] & E[X_n]E[X_2] & \cdots & E[X_n]^2 \end{pmatrix}.$$

Note that the first and second row are just the same. We have $\tilde{B}$ is singular, thus B is singular as well.

We claim that the matrix B is positive semi-definite. In fact, by deleting the same rows and columns of B , we always get a matrix of the form

$$\begin{pmatrix} a_1^2 & a_1a_2 & \cdots & a_1a_k \\ a_2a_1 & a_2^2 & \cdots & a_2a_k \\ \vdots & \vdots & \ddots & \vdots \\ a_ka_1 & a_ka_2 & \cdots & a_k^2 \end{pmatrix}$$

with $k \leq n$, which is proven to be singular when $k \geq 2$, or is just a 1×1 matrix with some non-negative entry. Thus we have that all principal minors of the symmetric matrix B are non-negative. B is thus positive semi-definite.

Finally, note we can write $A = B + D$ with

$$D = \begin{pmatrix} E[X_1^2] - E[X_1]^2 & & & \\ & E[X_2^2] - E[X_2]^2 & & \\ & & \ddots & \\ & & & E[X_n^2] - E[X_n]^2 \end{pmatrix},$$

which is a diagonal matrix. Since we also assumed $E[X_i^2] - E[X_i]^2 = \text{Var}[X_i] > 0$ for all $1 \leq i \leq n$, D is positive definite. For all $u \in \mathbb{R}^n \setminus \{0\}$ we have

$$u^T Au = u^T (B + D)u = u^T Bu + u^T Du > 0,$$

which shows A is positive definite and in particular invertible. □

We thus have a unique solution $\tilde{\pi}$ for equation (5.9). This $\tilde{\pi}$ is a local extremum of the function L_Q . Since we also know L_Q is convex, we finally get that $\tilde{\pi}$ is the unique global minimizer of L_Q .

Remark 5.7. One can also show that the Hessian matrix of L_Q at any point π is given by $2A$, which is positive-definite by Lemma 5.6. This is another way to show the convexity of L_Q .

Other requirements of Sion's Minimax Theorem can be shown exactly in the same way as in Section 5.1. Hence we also have the existence of an optimal strategy π^* in this situation.

6 The T-period, one-risky-asset case

In this chapter, we assume there is only one risky asset in the market. However, the model has T periods, which means that we have chance to change our shares of the assets at the beginning of every period. Purpose of this chapter is to find an optimal strategy π^* in this situation. The set of candidate measures is still assumed to be $\mathcal{Q}$ as in (2.3). The set of admissible strategies Π however, is no more the set of all $\mathbb{F}$ -predictable and $\mathbb{R}$ -valued random processes $\mathcal{A}(\mathbb{R})$. For any probability measure P , we define

$$\mathcal{A}_2^P(\mathbb{R}) = \{\pi \in \mathcal{A}(\mathbb{R}) : \pi_t(S_t - S_{t-1}) \in L^2(P), \forall t \in \{1, \dots, T\}\}.$$

The set of admissible strategies Π is now chosen to be

$$\Pi = \bigcap_{Q \in \mathcal{Q}} \mathcal{A}_2^Q(\mathbb{R}).$$

This choice of Π provides us square-integrability in this context, which is similar to the square integrability assumption of X we made in the previous chapters. The space of end values in this situation is defined to be

$$\mathcal{G} = \left\{ X_T^{0,\pi} = \sum_{t=1}^T \pi_t(S_t - S_{t-1}) : \pi \in \Pi \right\}.$$

By our definition of Π and the fact that $L^2(Q)$ is a vector space for all $Q \in \mathcal{Q}$, we have

$$\mathcal{G} \subseteq L^2(Q), \quad \forall Q \in \mathcal{Q}.$$

Proposition 6.1. *For all $Q \in \mathcal{Q}$, the subset $\mathcal{G}$ is actually a subspace of $L^2(Q)$.*

Proof. Let $G_1, G_2 \in \mathcal{G}$ and $\alpha \in \mathbb{R}$. We have

$$G_1 = X_T^{0,\pi_1} = \sum_{t=1}^T \pi_{1t}(S_t - S_{t-1})$$

and

$$G_2 = X_T^{0,\pi_2} = \sum_{t=1}^T \pi_{2t}(S_t - S_{t-1})$$

for some $\pi_1, \pi_2 \in \Pi$. We thus have

$$G_1 + G_2 = \sum_{t=1}^T (\pi_{1t} + \pi_{2t})(S_t - S_{t-1}) = \sum_{t=1}^T (\pi_1 + \pi_2)_t(S_t - S_{t-1}) = X_T^{0,\pi_1 + \pi_2}.$$

Since $\pi_1, \pi_2 \in \Pi$ and again by the fact $L^2(Q)$ is a vector space for all $Q \in \mathcal{Q}$, one can show $\pi_1 + \pi_2 \in \Pi$. This shows $G_1 + G_2 \in \mathcal{G}$. Similarly we have

$$\begin{aligned}\alpha G_1 &= \alpha X_T^{0, \pi_1} \\ &= \alpha \sum_{t=1}^T \pi_{1_t}(S_t - S_{t-1}) \\ &= \sum_{t=1}^T \alpha \pi_{1_t}(S_t - S_{t-1}) \\ &= \sum_{t=1}^T (\alpha \pi_1)_t(S_t - S_{t-1}) \\ &= X_T^{0, \alpha \pi_1},\end{aligned}$$

where $\alpha \pi_1 \in \Pi$ again because $L^2(Q)$ is a vector space for all $Q \in \mathcal{Q}$. This shows $\alpha G_1 \in \mathcal{G}$. Hence we have shown $\mathcal{G}$ is itself a vector space. $\square$

Let us assume as in chapter 5 that the space of end values $\mathcal{G}$ is closed in $L^2(Q)$ for all $Q \in \mathcal{Q}$. In addition, the payoff H is assumed to be square-integrable as usual, that is to say

$$E^Q[H^2] < \infty, \quad \forall Q \in \mathcal{Q}.$$

In the previous chapters, it is rather straight forward to define a topology on Π , since we can just choose the euclidean topology on $\mathbb{R}^n$. In this case it is difficult to do that, so we want to transform the problem on another set. Instead of observing the loss function defined in (2.2) directly, we observe

$$\begin{aligned}K : \mathcal{G} \times \mathcal{Q} &\rightarrow \mathbb{R} \\ (G, Q) &\mapsto E^Q[(\bar{H} - G)^2].\end{aligned}$$

On $\mathcal{Q}$ we can choose the same topology mentioned in chapter 4. Next we will discuss the choice of a topology on $\mathcal{G}$.

For any fixed $Q \in \mathcal{Q}$ we observe the function

$$\begin{aligned}K_Q : \mathcal{G} &\rightarrow \mathbb{R} \\ G &\mapsto K(G, Q).\end{aligned}$$

Since we want to apply Sion's Minimax Theorem (Theorem 3.5) to the function K , we want to choose a topology on $\mathcal{G}$ in such a way, that the function K_Q 's are simultaneously continuous for all $Q \in \mathcal{Q}$. Define $\tilde{Q} := \frac{1}{M} \sum_{m=1}^M Q_m$, where

$Q_1, Q_2, \dots, Q_M$ are the base models. For any measure μ , let $\|\cdot\|_{L^2(\mu)}$ be the L^2 norm on the space $L^2(\mu)$. We have the following proposition.

Proposition 6.2. *Let $\mathcal{G}$ be endowed with the norm $\|\cdot\|_{L^2(\tilde{Q})}$, then the function K_Q defined as above is continuous for any $Q \in \mathcal{Q}$.*

Proof. For all $G \in \mathcal{G}$ and $i \in \{1, \dots, M\}$ we have

$$\|G\|_{L^2(\tilde{Q})}^2 = \int_{\Omega} G^2 d\tilde{Q} = \frac{1}{M} \sum_{m=1}^M \int_{\Omega} G^2 dQ_m \geq \frac{1}{M} \int_{\Omega} G^2 dQ_i = \frac{1}{M} \|G\|_{L^2(Q_i)}^2.$$

Hence

$$\|G\|_{L^2(\tilde{Q})} \geq \frac{1}{\sqrt{M}} \|G\|_{L^2(Q_i)}, \quad \forall G \in \mathcal{G}, \quad \forall i \in \{1, \dots, M\}.$$

Let us be given some $\epsilon > 0$. Let $i \in \{1, \dots, n\}$ and $G_0 \in \mathcal{G}$ be arbitrary. We know that if we endow $\mathcal{G}$ with the norm $\|\cdot\|_{L^2(Q_i)}$, the function

$$\begin{aligned} K_{Q_i} : (\mathcal{G}, \|\cdot\|_{L^2(Q_i)}) &\rightarrow \mathbb{R} \\ G &\mapsto K(G, Q_i) = E^{Q_i}[(\bar{H} - G)^2] \end{aligned}$$

is continuous. This means we can find some $\delta > 0$ such that

$$\|G - G_0\|_{L^2(Q_i)} < \delta \Rightarrow |K_{Q_i}(G) - K_{Q_i}(G_0)| < \epsilon.$$

If $\|G - G_0\|_{L^2(\tilde{Q})} < \frac{\delta}{\sqrt{M}}$, then

$$\frac{1}{\sqrt{M}} \|G - G_0\|_{L^2(Q_i)} \leq \|G - G_0\|_{L^2(\tilde{Q})} < \frac{\delta}{\sqrt{M}}.$$

We thus have $\|G - G_0\|_{L^2(Q_i)} < \delta$. This implies

$$|K_{Q_i}(G) - K_{Q_i}(G_0)| < \epsilon.$$

We have just proven that for any $i \in \{1, \dots, M\}$ the function

$$\begin{aligned} K_{Q_i} : (\mathcal{G}, \|\cdot\|_{L^2(\tilde{Q})}) &\rightarrow \mathbb{R} \\ G &\mapsto K(G, Q_i) \end{aligned}$$

is continuous. For any arbitrary $Q \in \mathcal{Q}$, we can write

$$Q = \sum_{m=1}^M \alpha_m Q_m$$

for some $\alpha \in \mathbb{R}^M$. For any $G \in \mathcal{G}$ we have

$$K_Q(G) = \int_{\Omega} (\bar{H} - G)^2 dQ = \sum_{m=1}^M \alpha_m \int_{\Omega} (\bar{H} - G)^2 dQ_m = \sum_{m=1}^M \alpha_m K_{Q_m}(G).$$

We see that K_Q is just a linear combination of continuous functions on $(\mathcal{G}, \|\cdot\|_{L^2(\bar{Q})})$, thus K_Q is continuous on $(\mathcal{G}, \|\cdot\|_{L^2(\bar{Q})})$ as well. $\square$

By Remark 3.4 we have that for all $Q \in \mathcal{Q}$, the function K_Q is also lower semi-continuous.

Remark 6.3. We can define another topology on $\mathcal{G}$ which is independent of the structure of $\mathcal{Q}$. This will be introduced in Section 8 when we try generalizing the set $\mathcal{Q}$.

Proposition 6.4. *For any $Q \in \mathcal{Q}$, the function K_Q is convex, in particular it is quasi-convex.*

Proof. For any arbitrary $Q \in \mathcal{Q}$, $\lambda \in [0, 1]$ and $G, G' \in \mathcal{G}$ we have

$$\begin{aligned} K_Q(\lambda G + (1 - \lambda)G') &= E^Q \left[\left(\bar{H} - (\lambda G + (1 - \lambda)G') \right)^2 \right] \\ &= E^Q \left[\left(\lambda(\bar{H} - G) + (1 - \lambda)(\bar{H} - G') \right)^2 \right] \\ &\leq E^Q \left[\lambda(\bar{H} - G)^2 + (1 - \lambda)(\bar{H} - G')^2 \right] \\ &= \lambda E^Q[(\bar{H} - G)^2] + (1 - \lambda) E^Q[(\bar{H} - G')^2] \\ &= \lambda K_Q(G) + (1 - \lambda) K_Q(G'), \end{aligned}$$

which proves the convexity of K_Q . The quasi-convexity is again a direct result from Remark 3.2. $\square$

We can again use linear projection to prove the following proposition.

Proposition 6.5. *For any $Q \in \mathcal{Q}$, infimum of the function K_Q is attained.*

Proof. We fix some $Q \in \mathcal{Q}$. We know by assumption that $\mathcal{G}$ is a closed subspace of $L^2(Q)$. Thus we can observe the linear projection of $\bar{H}$ on the subspace $\mathcal{G}$. Denote it by $P(\bar{H})$. We have by the property of linear projection

$$E^Q[(\bar{H} - P(\bar{H}))^2] = \min_{G \in \mathcal{G}} E^Q[(\bar{H} - G)^2],$$

which is just the desired result. $\square$

We are finished with the study of the function K_Q . Let V be defined as in 4.1. For any $G \in \mathcal{G}$, we observe now the function

$$\begin{aligned}\tilde{K}_G : V &\rightarrow \mathbb{R} \\ Q &\mapsto \int_{\Omega} (\bar{H} - G)^2 dQ.\end{aligned}$$

One can show that for any $G \in \mathcal{G}$, $\tilde{K}_G$ is a linear map between V and $\mathbb{R}$. In fact, we fix some $G \in \mathcal{G}$. For all $Q_1, Q_2 \in V$ and $\alpha, \beta \in \mathbb{R}$ we have then

$$\begin{aligned}\tilde{K}_G(\alpha Q_1 + \beta Q_2) &= \int_{\Omega} (\bar{H} - G)^2 d(\alpha Q_1 + \beta Q_2) \\ &= \alpha \int_{\Omega} (\bar{H} - G)^2 dQ_1 + \beta \int_{\Omega} (\bar{H} - G)^2 dQ_2 \\ &= \alpha \tilde{K}_G(Q_1) + \beta \tilde{K}_G(Q_2).\end{aligned}$$

In particular, the above proposition shows that the function $\tilde{K}_G$ is continuous for all $G \in \mathcal{G}$, since a linear function over a finite dimensional vector space is always continuous. We observe the restriction of $\tilde{K}_G$ on $\mathcal{Q}$

$$\begin{aligned}K_G : \mathcal{Q} &\rightarrow \mathbb{R} \\ Q &\mapsto \int_{\Omega} (\bar{H} - G)^2 dQ = E^Q[(\bar{H} - G)^2] = K(G, Q).\end{aligned}$$

Proposition 6.6. *For any $G \in \mathcal{G}$, K_G is upper semi-continuous and quasi-concave.*

Proof. We fix an arbitrary $G \in \mathcal{G}$. As restriction of a continuous function, K_G is clearly also continuous, hence upper semi-continuous by Remark 3.4. Furthermore, for any $\lambda \in [0, 1]$ and any $Q_1, Q_2 \in \mathcal{Q}$, we have

$$\begin{aligned}K_G(\lambda Q_1 + (1 - \lambda)Q_2) &= E^{\lambda Q_1 + (1 - \lambda)Q_2}[(\bar{H} - G)^2] \\ &= \lambda E^{Q_1}[(\bar{H} - G)^2] + (1 - \lambda) E^{Q_2}[(\bar{H} - G)^2] \\ &= \lambda K_G(Q_1) + (1 - \lambda) K_G(Q_2) \\ &\geq \min\{K_G(Q_1), K_G(Q_2)\}.\end{aligned}$$

This shows us that K_G is quasi-concave. □

We are now in the situation to prove the main theorem of this section.

Theorem 6.7. *In the above setting, there exists an optimal strategy $\pi^* \in \Pi$ such that*

$$L^* = \min_{\pi \in \Pi} \max_{Q \in \mathcal{Q}} L(\pi, Q) = \max_{Q \in \mathcal{Q}} L(\pi^*, Q),$$

where L is defined in (2.2) as usual.

Proof. We observe the function

$$\begin{aligned} K : \mathcal{G} \times \mathcal{Q} &\rightarrow \mathbb{R} \\ (G, Q) &\mapsto E^Q[(\bar{H} - G)^2], \end{aligned}$$

where $\mathcal{G}$ endowed with the norm $\|\cdot\|_{L^2(\tilde{Q})}$ is a convex subset of the linear topological space $L^2(\tilde{Q})$ and $\mathcal{Q}$ is a compact and convex subset of the linear topological space V . The function K is real-valued and satisfies

$K(\cdot, Q)$ is quasi-convex and lower semi-continuous on $\mathcal{G}$, $\forall Q \in \mathcal{Q}$, and

$K(G, \cdot)$ is quasi-concave and upper semi-continuous on $\mathcal{Q}$, $\forall G \in \mathcal{G}$.

We can then apply Sion's minimax theorem (Theorem 3.5) to show

$$\max_{Q \in \mathcal{Q}} \inf_{G \in \mathcal{G}} K(G, Q) = \inf_{G \in \mathcal{G}} \max_{Q \in \mathcal{Q}} K(G, Q).$$

Furthermore, we have also shown that for any fixed $Q \in \mathcal{Q}$, $\inf_{G \in \mathcal{G}} K(G, Q)$ is attained. We can then apply Remark 3.6 to show the existence of some $G^* \in \mathcal{G}$, such that

$$\max_{Q \in \mathcal{Q}} K(G^*, Q) = \min_{G \in \mathcal{G}} \max_{Q \in \mathcal{Q}} K(G, Q). \quad (6.1)$$

By definition of the set $\mathcal{G}$, we can find some $\pi^* \in \Pi$ such that

$$G^* = X_T^{0, \pi^*}.$$

We claim

$$\max_{Q \in \mathcal{Q}} L(\pi^*, Q) = \min_{\pi \in \Pi} \max_{Q \in \mathcal{Q}} L(\pi, Q) = L^*. \quad (6.2)$$

In fact, assume by contradiction there is some $\tilde{\pi} \in \Pi$ such that

$$\max_{Q \in \mathcal{Q}} L(\tilde{\pi}, Q) < \max_{Q \in \mathcal{Q}} L(\pi^*, Q).$$

Define

$$\tilde{G} := X_T^{0, \tilde{\pi}} \in \mathcal{G}.$$

By (2.2), we have

$$L(\pi^*, Q) = E^Q[(H - X_T^{c, \pi^*})^2] = E^Q[(\bar{H} - X_T^{0, \pi^*})^2] = E^Q[(\bar{H} - G^*)^2] = K(G^*, Q)$$

for all $Q \in \mathcal{Q}$. Similarly, we have

$$L(\tilde{\pi}, Q) = K(\tilde{G}, Q)$$

for all $Q \in \mathcal{Q}$. We thus have

$$\max_{Q \in \mathcal{Q}} K(\tilde{G}, Q) = \max_{Q \in \mathcal{Q}} L(\tilde{\pi}, Q) < \max_{Q \in \mathcal{Q}} L(\pi^*, Q) = \max_{Q \in \mathcal{Q}} K(G^*, Q),$$

which contradicts (6.1). This proves that Equation (6.2) is true, which is the desired result. $\square$

7 The T-period, n-risky-asset case

In this chapter, we want to see the existence of an optimal strategy in a general T-period, n-risky-asset model. The candidate set of measures $\mathcal{Q}$ is still assumed to be the same as in section 4. The procedure here is very similar to the one we introduced in section 6 and can be seen as a generalisation of the concept before.

Suppose P is a measure. Let us define

$$\mathcal{A}_2^P(\mathbb{R}^n) = \left\{ \pi \in \mathcal{A}(\mathbb{R}^n) : \pi_t \cdot (S_t - S_{t-1}) \in L^2(P), \forall t \in \{1, \dots, T\} \right\}.$$

The set of admissible strategies Π is chosen to be

$$\Pi = \bigcap_{Q \in \mathcal{Q}} \mathcal{A}_2^Q(\mathbb{R}^n).$$

We define the space of end values to be

$$\mathcal{G} = \left\{ X_T^{0,\pi} = \sum_{t=1}^T \pi_t \cdot (S_t - S_{t-1}) : \pi \in \Pi \right\}.$$

As in the previous chapters, we also have

$$\mathcal{G} \subseteq L^2(Q) \quad \forall Q \in \mathcal{Q}$$

in this case.

Proposition 7.1. *For all $Q \in \mathcal{Q}$, the subset $\mathcal{G}$ is actually a subspace of $L^2(Q)$.*

Proof. Let $G_1, G_2 \in \mathcal{G}$ and $\alpha \in \mathbb{R}$. We have

$$G_1 = X_T^{0,\pi_1} = \sum_{t=1}^T \pi_{1t} \cdot (S_t - S_{t-1})$$

and

$$G_2 = X_T^{0,\pi_2} = \sum_{t=1}^T \pi_{2t} \cdot (S_t - S_{t-1})$$

for some $\pi_1, \pi_2 \in \Pi$. We thus have

$$G_1 + G_2 = \sum_{t=1}^T (\pi_{1t} + \pi_{2t}) \cdot (S_t - S_{t-1}) = \sum_{t=1}^T (\pi_1 + \pi_2)_t \cdot (S_t - S_{t-1}) = X_T^{0,\pi_1+\pi_2}.$$

Since $\pi_1, \pi_2 \in \Pi$ and again by the fact $L^2(Q)$ is a vector space for all $Q \in \mathcal{Q}$, one

can show $\pi_1 + \pi_2 \in \Pi$. This shows $G_1 + G_2 \in \mathcal{G}$. Similarly we have

$$\begin{aligned}
\alpha G_1 &= \alpha X_T^{0, \pi_1} \\
&= \alpha \sum_{t=1}^T \pi_{1t} \cdot (S_t - S_{t-1}) \\
&= \sum_{t=1}^T \alpha (\pi_{1t} \cdot (S_t - S_{t-1})) \\
&= \sum_{t=1}^T (\alpha \pi_1)_t \cdot (S_t - S_{t-1}) \\
&= X_T^{0, \alpha \pi_1},
\end{aligned}$$

where $\alpha \pi_1 \in \Pi$ again because $L^2(Q)$ is a vector space for all $Q \in \mathcal{Q}$. This shows $\alpha G_1 \in \mathcal{G}$. Hence we have shown $\mathcal{G}$ is itself a vector space. $\square$

The subspace $\mathcal{G}$ is again assumed to be closed in $L^2(Q)$ for all $Q \in \mathcal{Q}$. Let us assume in addition

$$E^Q[H^2] < \infty, \quad \forall Q \in \mathcal{Q}$$

just like before.

We again observe functions $K, K_Q, \tilde{K}_G$ and K_G with exactly the same definitions as in chapter 6. The results from Proposition 6.2 to 6.6 can be also applied in this situation and be proven exactly in the same way. The main theorem in this chapter is then

Theorem 7.2. *In the above setting, there exists an optimal strategy $\pi^* \in \Pi$ such that*

$$L^* = \min_{\pi \in \Pi} \max_{Q \in \mathcal{Q}} L(\pi, Q) = \max_{Q \in \mathcal{Q}} L(\pi^*, Q),$$

where L is defined in (2.2) as usual.

Proof. The proof is exactly the same as the proof of Theorem 6.7. In fact, we can just copy the proof without changing a word. $\square$

8 Generalisation of the candidate set $\mathcal{Q}$

In this last chapter, we study a T-period, n-risky-asset model. However, the set of candidate measures is no more the set defined in Equation (2.3). We want to make some further assumptions and try concluding the existence result in this setting.

Before starting the discussion, let us recall the definition of *Borel σ -algebra*.

Definition 8.1. Suppose X is a topological space. The smallest σ -algebra containing the topology of X is called the *Borel σ -algebra of X* . The Borel σ -algebra of X is denoted by $\mathcal{B}_X$ and elements of $\mathcal{B}_X$ are called *Borel Sets*.

For sake of simplicity, we write $ca(X) = ca(\mathcal{B}_X)$ if X is a topological space. Recall that $ca(\mathcal{B}_X)$ is the set of all signed measures on $\mathcal{B}_X$.

We assume for now that the set of states Ω is a topological space and the σ -algebra $\mathcal{F}$ of all events we want to observe is $\mathcal{B}_\Omega$. Since we do not know the structure of $\mathcal{Q}$ in this case, we cannot observe the vector space V as defined in 4.1. We also need another topology in this case. The following definition of *narrow topology* is referenced from Bell [4, Section 5].

Definition 8.2. Suppose that X is a topological space. Denote the set of all bounded and continuous functions $f : X \rightarrow \mathbb{R}$ by $C_b(X)$. The *narrow topology* on $ca(X)$ is the coarsest topology such that for each $f \in C_b(X)$, the map $\mu \mapsto \int_X f d\mu$ is continuous $ca(X) \rightarrow \mathbb{R}$. The narrow topology is also called the *weak- * topology*.

In our setting, $\mathcal{Q}$ is a general subset of $ca(\Omega)$, which we assume to be convex and compact. Note also that $C_b(\Omega) \subseteq L^2(Q)$ for all $Q \in \mathcal{Q}$. In fact, if $|f| \leq a$ for some $a \in \mathbb{R}$, then $f^2 \leq a^2$. We thus have

$$E^Q[f^2] \leq a^2 < \infty,$$

for any $Q \in \mathcal{Q}$, which is the desired result. Note also $C_b(\Omega)$ is itself a vector space. We define the set of admissible strategies Π to be

$$\Pi = \left\{ \pi \in \mathcal{A}(\mathbb{R}^n) : \pi_t \cdot (S_t - S_{t-1}) \in C_b(\Omega), \forall t \in \{1, \dots, T\} \right\}$$

and again observe the set of end values

$$\mathcal{G} = \left\{ X_T^{0,\pi} = \sum_{t=1}^T \pi_t \cdot (S_t - S_{t-1}) : \pi \in \Pi \right\}.$$

We have a similar result as in the previous chapters.

Proposition 8.3. *The set $\mathcal{G}$ is actually a subspace of $C_b(\Omega)$.*

Proof. By the fact that $C_b(\Omega)$ is a vector space and by definition elements of $\mathcal{G}$ are just sum of functions in $C_b(\Omega)$, we have $\mathcal{G}$ is a subset of $C_b(\Omega)$. Let $G_1, G_2 \in \mathcal{G}$ and $\alpha \in \mathbb{R}$. We have

$$G_1 = X_T^{0, \pi_1} = \sum_{t=1}^T \pi_{1_t} \cdot (S_t - S_{t-1})$$

and

$$G_2 = X_T^{0, \pi_2} = \sum_{t=1}^T \pi_{2_t} \cdot (S_t - S_{t-1})$$

for some $\pi_1, \pi_2 \in \Pi$. We thus have

$$G_1 + G_2 = \sum_{t=1}^T (\pi_{1_t} + \pi_{2_t}) \cdot (S_t - S_{t-1}) = \sum_{t=1}^T (\pi_1 + \pi_2)_t \cdot (S_t - S_{t-1}) = X_T^{0, \pi_1 + \pi_2}.$$

Since $\pi_1, \pi_2 \in \Pi$ and by the fact $C_b(\Omega)$ is a vector space, one can show $\pi_1 + \pi_2 \in \Pi$. This shows $G_1 + G_2 \in \mathcal{G}$. Similarly we have

$$\begin{aligned} \alpha G_1 &= \alpha X_T^{0, \pi_1} \\ &= \alpha \sum_{t=1}^T \pi_{1_t} \cdot (S_t - S_{t-1}) \\ &= \sum_{t=1}^T \alpha (\pi_{1_t} \cdot (S_t - S_{t-1})) \\ &= \sum_{t=1}^T (\alpha \pi_1)_t \cdot (S_t - S_{t-1}) \\ &= X_T^{0, \alpha \pi_1}, \end{aligned}$$

where $\alpha \pi_1 \in \Pi$ again because $C_n(\Omega)$ is a vector space. This shows $\alpha G_1 \in \mathcal{G}$. Hence we have shown $\mathcal{G}$ is itself a vector space. $\square$

We again assume that $\mathcal{G}$ is a closed subspace of $L^2(Q)$, for all $Q \in \mathcal{Q}$. Furthermore, let us assume

$$H \in C_b(\Omega). \tag{8.1}$$

We again observe the function

$$\begin{aligned} K : \mathcal{G} \times \mathcal{Q} &\rightarrow \mathbb{R} \\ (G, Q) &\mapsto E^Q[(\bar{H} - G)^2], \end{aligned}$$

where $\mathcal{G}$ is a subset of $C_b(\Omega)$, which is endowed with the topology of pointwise convergence, and $\mathcal{Q}$ is a subset of $ca(\Omega)$, which is endowed with the narrow topology. For any fixed $Q \in \mathcal{Q}$, we also observe

$$\begin{aligned} K_Q : \mathcal{G} &\rightarrow \mathbb{R} \\ G &\mapsto K(G, Q). \end{aligned}$$

Proposition 8.4. *With the chosen topology on $\mathcal{G}$, K_Q is lower semi-continuous for all $Q \in \mathcal{Q}$.*

Proof. We fix an arbitrary $Q \in \mathcal{Q}$ and construct the following sequence of functions

$$K_{Q,j}(G) = E^Q[(\bar{H} - G)^2 \wedge j], \quad j \in \mathbb{N}.$$

where $\bar{H} = H - c$ as usual and $a \wedge b = \min\{a, b\}$. Clearly, we have $L_{Q,j}$ is increasing in j for all $G \in \mathcal{G}$. In addition

$$\lim_{j \rightarrow \infty} L_{Q,j}(G) = L_Q(G), \quad \forall G \in \mathcal{G}.$$

We now want to show that $L_{Q,j}$ is continuous $\forall j \in \mathbb{N}$, then we can conclude L_Q is lower semi-continuous. We fix an arbitrary $j \in \mathbb{N}$. Assume $G_n \rightarrow G$ in $\mathcal{G}$ for some sequence $(G_n)_n$, that is to say

$$G_n(\omega) \rightarrow G(\omega), \quad \forall \omega \in \Omega.$$

This implies

$$(\bar{H} - G_n)^2 \wedge j \rightarrow (\bar{H} - G)^2 \wedge j$$

pointwise. Since for all $n \in \mathbb{N}$

$$(\bar{H} - G_n)^2 \wedge j \leq j$$

is bounded from above by j , we can use the dominated convergence theorem to conclude

$$L_{Q,j}(G_n) = E^Q[(\bar{H} - G_n)^2 \wedge j] \rightarrow E^Q[(\bar{H} - G)^2 \wedge j] = L_{Q,j}(G).$$

This shows us the continuity of $L_{Q,j}$ and finishes the proof. $\square$

In addition, for any $Q \in \mathcal{Q}$ the function K_Q is convex and attains its infimum on $\mathcal{G}$ under our assumption. Proofs of these two results are just identical with those

in section 6.

For any fixed $G \in \mathcal{G}$, we observe the function

$$\begin{aligned} K_G : \mathcal{Q} &\rightarrow \mathbb{R} \\ Q &\mapsto K(G, Q). \end{aligned}$$

In the setting of this chapter, we can no longer find a finite dimensional vector space V , which contains the candidate set $\mathcal{Q}$. Thus, the linearity of K_G does not provides us the continuity. Some further assumptions we made gives us the continuity result in this situation.

Proposition 8.5. *In the above setting, function K_G is continuous for all $G \in \mathcal{G}$.*

Proof. We fix some $G \in \mathcal{G}$ and observe

$$\begin{aligned} \tilde{K}_G : ca(\Omega) &\rightarrow \mathbb{R} \\ Q &\mapsto \int_{\Omega} (\bar{H} - G)^2 dQ, \end{aligned}$$

where the set $ca(\Omega)$ is endowed with the narrow topology. By (8.1) and Proposition 8.3 we get

$$(\bar{H} - G)^2 \in C_b(\Omega).$$

By definition of narrow topology, the function $\tilde{K}_G$ is continuous. Thus the restriction of $\tilde{K}_G$ on $\mathcal{Q}$, which is exactly K_G , is continuous as well. This finishes the proof. $\square$

The fact that K_G is quasi-concave can be shown in the same way as before. We thus have the existence result in this context.

Theorem 8.6. *In the above setting, there exists an optimal strategy $\pi^* \in \Pi$ such that*

$$L^* = \min_{\pi \in \Pi} \max_{Q \in \mathcal{Q}} L(\pi, Q) = \max_{Q \in \mathcal{Q}} L(\pi^*, Q).$$

Proof. We observe the function

$$\begin{aligned} K : \mathcal{G} \times \mathcal{Q} &\rightarrow \mathbb{R} \\ (G, Q) &\mapsto E^Q[(\bar{H} - G)^2], \end{aligned}$$

where $\mathcal{G}$ is a convex subset of the linear topological space $C_b(\Omega)$, which is endowed with the topology of pointwise convergence. The set $\mathcal{Q}$, which is assumed to be

compact and convex, is a subset of the linear topological space $ca(\Omega)$, which is endowed with the narrow topology. The function K is real-valued and satisfies

$K(\cdot, Q)$ is quasi-convex and lower semi-continuous on $\mathcal{G}$, $\forall Q \in \mathcal{Q}$, and

$K(G, \cdot)$ is quasi-concave and upper semi-continuous on $\mathcal{Q}$, $\forall G \in \mathcal{G}$.

We can then apply Sion's minimax theorem (Theorem 3.5) to show

$$\max_{Q \in \mathcal{Q}} \inf_{G \in \mathcal{G}} K(G, Q) = \inf_{G \in \mathcal{G}} \max_{Q \in \mathcal{Q}} K(G, Q).$$

Furthermore, for any fixed $Q \in \mathcal{Q}$, $\inf_{G \in \mathcal{G}} K(G, Q)$ is attained. We can then apply Remark 3.6 to show the existence of some $G^* \in \mathcal{G}$, such that

$$\max_{Q \in \mathcal{Q}} K(G^*, Q) = \min_{G \in \mathcal{G}} \max_{Q \in \mathcal{Q}} K(G, Q).$$

Rest of the proof is exactly same with the one of Theorem 6.7. This shows us the existence of an optimal strategy $\pi^* \in \Pi$ such that

$$L^* = \min_{\pi \in \Pi} \max_{Q \in \mathcal{Q}} L(\pi, Q) = \max_{Q \in \mathcal{Q}} L(\pi^*, Q).$$

□

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